

Program 2:-Maven Project with Guava and Commons-IO Dependencies

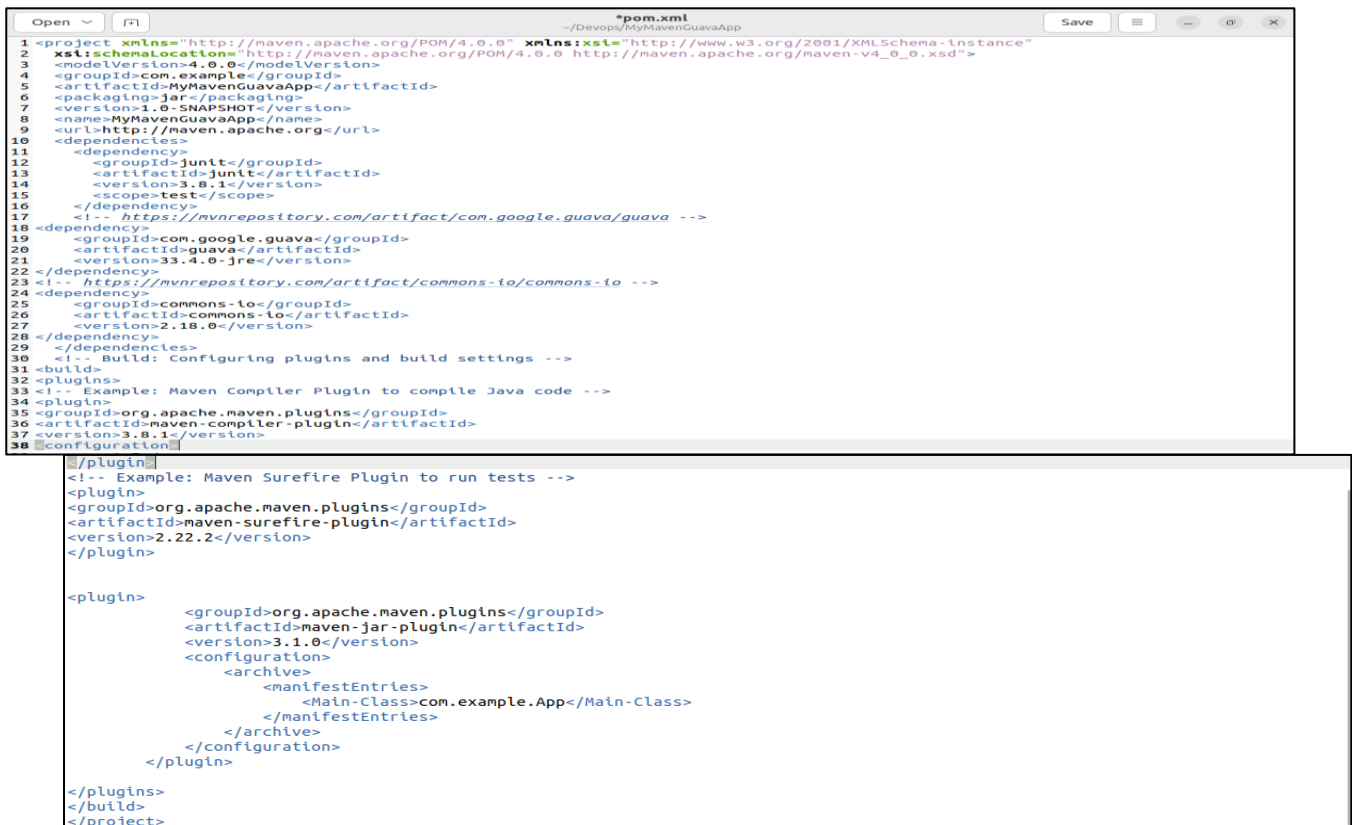
Create a Maven Project:

mvn archetype: generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
rakshith@rakshith-VirtualBox:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO]
[INFO] Parameter: basedir, Value: /home/rakshith/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenGuavaApp
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/rakshith/devops/MyMavenGuavaApp
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.404 s
[INFO] Finished at: 2026-03-19T13:32:06+05:30
[INFO]
rakshith@rakshith-VirtualBox:~/devops$
```

Change directory using the command: cd MyMavenGuavaApp

After changing that Edit pom.xml: gedit pom.xml

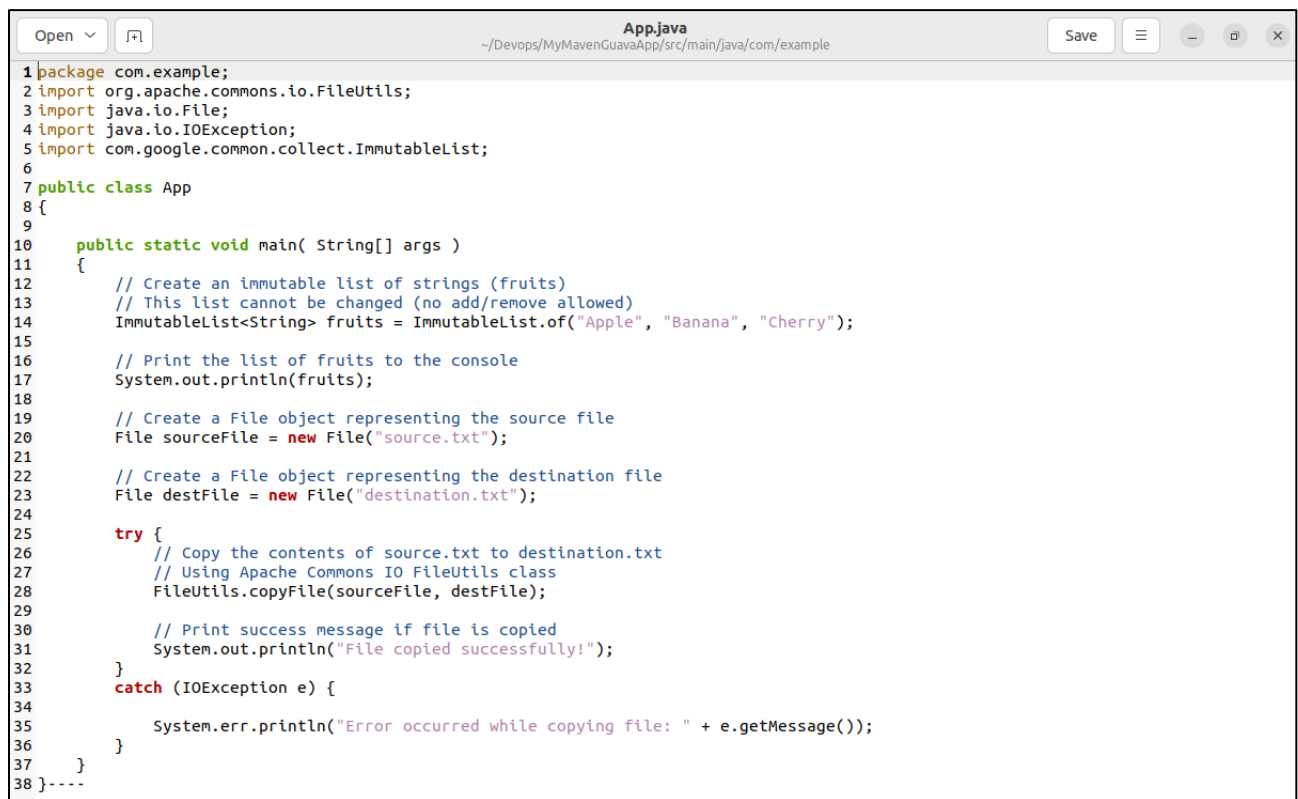


```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.example</groupId>
    <artifactId>MyMavenGuavaApp</artifactId>
    <packaging>jar</packaging>
    <version>1.0-SNAPSHOT</version>
    <name>MyMavenGuavaApp</name>
    <url>http://maven.apache.org</url>
    <dependencies>
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>3.8.1</version>
            <scope>test</scope>
        </dependency>
        <!-- https://mvnrepository.com/artifact/com.google.guava/guava -->
        <dependency>
            <groupId>com.google.guava</groupId>
            <artifactId>guava</artifactId>
            <version>33.4.0-jre</version>
        </dependency>
        <!-- https://mvnrepository.com/artifact/commons-io/commons-io -->
        <dependency>
            <groupId>commons-io</groupId>
            <artifactId>commons-io</artifactId>
            <version>2.18.0</version>
        </dependency>
    </dependencies>
    <!-- Build: Configuring plugins and build settings -->
    <build>
        <plugins>
            <!-- Example: Maven Compiler Plugin to compile Java code -->
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.8.1</version>
            </plugin>
            <!-- Example: Maven Surefire Plugin to run tests -->
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>2.22.2</version>
            </plugin>
            <!-- Example: Maven Jar Plugin to create the jar file -->
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-jar-plugin</artifactId>
                <version>3.1.0</version>
                <configuration>
                    <archive>
                        <manifestEntries>
                            <Main-Class>com.example.App</Main-Class>
                        </manifestEntries>
                    </archive>
                </configuration>
            </plugin>
        </plugins>
    </build>
</project>
```

Modify App.java:

- Navigate to cd src/main/java/com/example/

- Then perform gedit App.java



```
1 package com.example;
2 import org.apache.commons.io.FileUtils;
3 import java.io.File;
4 import java.io.IOException;
5 import com.google.common.collect.ImmutableList;
6
7 public class App
8 {
9
10     public static void main( String[] args )
11     {
12         // Create an immutable list of strings (fruits)
13         // This list cannot be changed (no add/remove allowed)
14         ImmutableList<String> fruits = ImmutableList.of("Apple", "Banana", "Cherry");
15
16         // Print the list of fruits to the console
17         System.out.println(fruits);
18
19         // Create a File object representing the source file
20         File sourceFile = new File("source.txt");
21
22         // Create a File object representing the destination file
23         File destFile = new File("destination.txt");
24
25         try {
26             // Copy the contents of source.txt to destination.txt
27             // Using Apache Commons IO FileUtils class
28             FileUtils.copyFile(sourceFile, destFile);
29
30             // Print success message if file is copied
31             System.out.println("File copied successfully!");
32         }
33         catch (IOException e) {
34
35             System.err.println("Error occurred while copying file: " + e.getMessage());
36         }
37     }
38 }----
```

Build the Project using the command mvn clean install:

```
(base) rakshith@rakshith-VirtualBox:/devopss6d/MyMavenGuavaApp$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenGuavaApp >-----
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MyMavenGuavaApp ---
[INFO] Deleting /home/rakshith//devopss6d/MyMavenGuavaApp/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform depe
[INFO] skip non existing resourceDirectory /home/rakshith//devopss6d/MyMavenGuavaApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyMavenGuavaApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/rakshith//devopss6d/MyMavenGuavaApp/target/classes
[INFO]
[INFO] --- maven-resources-plugin:3.6:testResources (default-testResources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform depe
[INFO] skip non existing resourceDirectory /home/rakshith//devopss6d/MyMavenGuavaApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:testCompile (default-testCompile) @ MyMavenGuavaApp ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to /home/rakshith//devopss6d/MyMavenGuavaApp/target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenGuavaApp ---
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.008 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- maven-jar-plugin:3.1.0:jar (default-jar) @ MyMavenGuavaApp ---
[INFO] Building jar: /home/rakshith//devopss6d/MyMavenGuavaApp/target/MyMavenGuavaApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenGuavaApp ---
[INFO] Installing /home/rakshith//devopss6d/MyMavenGuavaApp/target/MyMavenGuavaApp-1.0-SNAPSHOT.jar to /ho
[INFO] Installing /home/rakshith//devopss6d/MyMavenGuavaApp/pom.xml to /home/rakshith//.m2/repository/com
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.418 s
[INFO] Finished at: 2026-05-17T18:03:27Z
[INFO]
[INFO] -----
```

This compiles the project and creates a JAR in the target/ folder.

Run the Application:

use the following command :

`mvn exec:java -Dexec.mainClass="com.example.App"`

```
(base) rakshith@rakshith-VirtualBox:~/devops6d/MyMavenGuavaApp$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyMavenGuavaApp >-----
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenGuavaApp ---
[INFO] Apple, Banana, Cherry
[INFO] File copied successfully!
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.657 s
[INFO] Finished at: 2026-03-17T18:03:53Z
[INFO]
[INFO] -----
(base) rakshith@rakshith-VirtualBox:~/devops6d/MyMavenGuavaApp$
```

Now, Create a Sample File to copy the content to destination file



Now Run the command `mvn exec:java -Dexec.mainClass="com.example.App"` again

```
(base) rakshith@rakshith-VirtualBox:~/devops6d/MyMavenGuavaApp$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO] ----- < com.example:MyMavenGuavaApp > -----
[INFO] Building MyMavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ] -----
[INFO] --- exec-maven-plugin:3.6.3:java (default-cli) @ MyMavenGuavaApp ---
[Apple, Banana, Cherry]
File copied successfully!
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.657 s
[INFO] Finished at: 2026-03-17T18:03:53Z
[INFO] -----
(base) rakshith@rakshith-VirtualBox:~/devops6d/MyMavenGuavaApp$ █
```

The output shows file copied successfully.

If u want contents present in the directory can be checked using the command: ls

```
rakshith@rakshith-VirtualBox:~/devops/MyMavenGuavaApp$ ls
pom.xml source.txt src target
rakshith@rakshith-VirtualBox:~/devops/MyMavenGuavaApp$ █
```

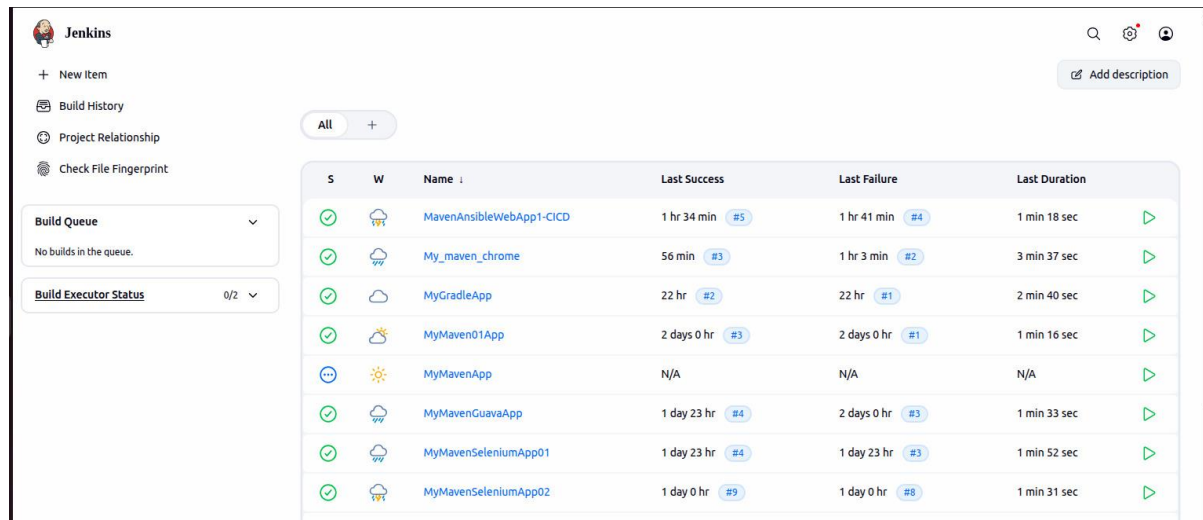
And in destination file the content will be same as that of the source file



A screenshot of a text editor window titled "destination.txt" with the path "~/Devops/MyMavenGuavaApp". The window contains a single line of text: "1 hoooooooooooo". The editor has a menu bar with "Open", "Save", and a hamburger menu icon, and standard window control buttons (minimize, maximize, close) on the right.

Jenkins Pipeline for MyMavenGuava:

Step 1:Open Jenkins Dashboard to create a new item



Step 2: Click on New Item in order for creating an New Item and thus give the name as “MyMavenGuavaApp” and Item type as Pipeline and Create it

New Item

Enter an item name

MyMavenGuavaApp

Select an item type



Pipeline

Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.



Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.



Maven project

Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.



Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



Folder

OK

Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/rakshith31-hub/MyMavenGuavaApp.git

Credentials ?

GitHub Credential

+ Add

Advanced

Save Apply

Step 4: Now in Terminal, go to directory created for MyMavenGuavaApp and create an new file called “Jenkinsfile”

Paste the code inside Jenkinsfile

Enter following code into jenkins file

```
pipeline
{
    agent any // Use any available agent

    tools {
        maven 'Maven' // Ensure this matches the name configured in Jenkins
    }

    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url:
'https://github.com/rakshith31-hub/Mymavenapp.git'
            }
        }
    }
}
```

```
stage('Build')
{
    steps {
        sh 'mvn clean package' // Run Maven build
    }
}

stage('Test')
{
    steps {
        sh 'mvn test' // Run unit tests
    }
}

stage('Run Application') {
    steps {
        // Start the JAR application
        sh 'java -jar target/MyMavenApp-1.0-SNAPSHOT.jar'
    }
}

}

post {
    success {
        echo 'Build and deployment successful!'
    }
    failure {
        echo 'Build failed!'
    }
}
}
```

Step 5: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 6: Now we need to Push changes and commit by pushing the necessary “Jenkinsfile” to the respective Repo

The screenshot shows the GitHub repository page for 'MyMavenGuavaApp' by user 'rakshith31-hub'. The repository is public and has 6 commits. The file list includes 'src', 'target', 'Jenkinsfile', 'destination.txt', 'pom.xml', 'souce.txt', and 'source.txt'. The 'Jenkinsfile' was updated 2 days ago. The 'About' section indicates no description, website, or topics are provided. The 'Releases' section shows no releases published. The 'Packages' section shows no packages published.

Step 7: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.

Jenkins

All

▼

MyMavenGuavaApp

🔍

⚙️

👤

Status

</> Changes

▶ Build Now

⚙️ Configure

🗑️ Delete Pipeline

🔍 Full Stage View

📁 Stages

✎ Rename

❓ Pipeline Syntax

Builds >

🔍 Filter

13 April 2026

🟢 #4

14:14

▼

🔴 #3

14:07

▼

🔴 #2

14:04

▼

🔴 #1

14:01

▼

🟢 MyMavenGuavaApp

Add description

Stage View

Average stage times:
(full run time: ~1min 33s)

🟢 #4

Apr 13
14:14

2
commits

Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Test	Declarative: Post Actions
3s	603ms	2s	42s	28s	525ms
3s	603ms	2s	42s	28s	525ms

Permalinks

- [Last build \(#4\), 2 days 3 hr ago](#)
- [Last stable build \(#4\), 2 days 3 hr ago](#)
- [Last successful build \(#4\), 2 days 3 hr ago](#)
- [Last failed build \(#3\), 2 days 3 hr ago](#)
- [Last unsuccessful build \(#3\), 2 days 3 hr ago](#)
- [Last completed build \(#4\), 2 days 3 hr ago](#)

REST API

Jenkins 2.558